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High Performance Processors and Systems

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Name
Last Name

Problem 1 (25%)	
Problem 2 (25%)	
Question 1 (25%)	
Question 2 (25%)	
Total (16 points)	
Extra (4 points)	
Total (20 points)	

Problem 1

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 100, R1 is set to 0)

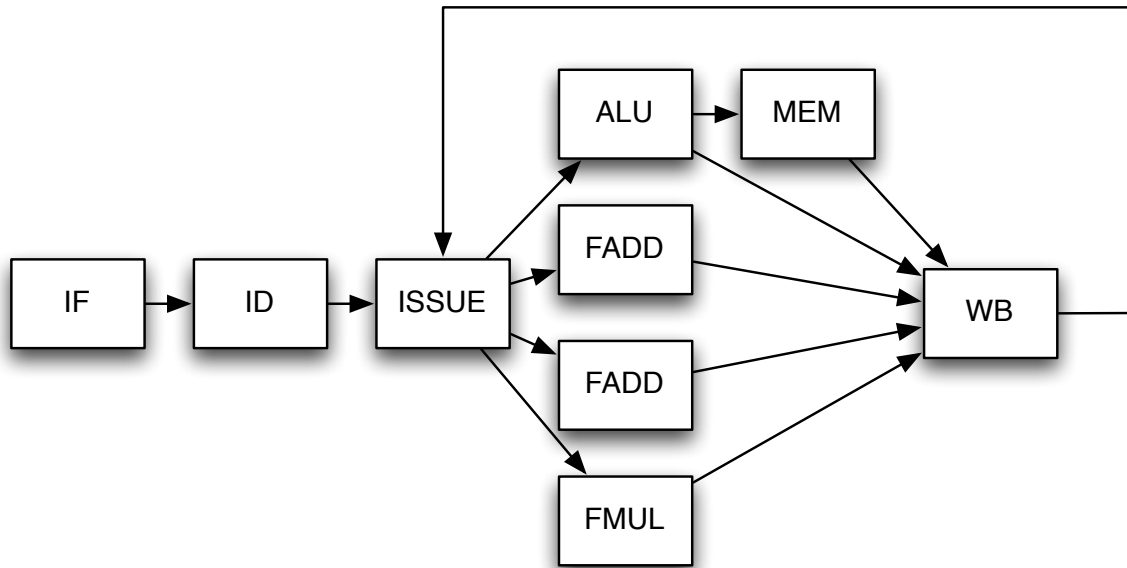
```
LOOP:    LD      F1  0    R0
         ADDD    F2  F1  F1
         ADDI    R1  R1  100
LOOP2:   MULTD   F2  F2  F1
         SUBI    R1  R1  1
         BNEZ    R1  LOOP2
         SUBI    R0  R0  1
         BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 2 cycles (includes time in ALU)
- Floating-point add instructions take 3 cycles
- Floating-point multiply instructions take 4 cycles
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

- **I** - When an instruction enters the issue stage
- **E** - When an instruction starts execution (enters FU)
- **C** - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

I ₁	L.D	F0, 0(R0)
I ₂	MUL.D	F0, F1, F1
I ₃	ADD.D	F2, F1, F1
I ₄	ADD.D	F4, F3, F3
I ₅	ADDI	R1, R3, 8
I ₆	ADD.D	F6, F0, F0

Problem 1.A

Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

[illegible]

Question 1

1. Explain the main characteristics of the scoreboard algorithm. Be specific.
2. How WAW and WAR are treated in the algorithm?

Question 2

Present the different types of dependences among instructions that can occur and relate them to the possible hazards that may occur in a pipeline.

Extra

Assume that a given 5 stage pipeline MIPS architecture is working with a clock cycle equal to 2 ns. Using the MIPS code shown below,

I1: add \$s3, \$s2, 2
I2: sub \$s4, \$s3, \$s1
I3: add \$s5, \$s4, \$s1
I4: lw \$s6, 4(\$s4)
I5: sub \$s7, \$s4, \$s6

- Draw the pipeline schema showing all the RAW data conflicts. Solve the conflicts without using rescheduling and path forwarding.
- Reschedule the instructions to reduce the stalls; Draw the pipeline schema showing all the RAW data conflicts.
- The forwarding paths have been included in the pipeline. Start from the code in (a) and draw the pipeline schema showing all the forwarding paths that have to be used to solve the conflicts.

Answer

- a) Show all the RAW data conflicts

Istr	CK1	CK2	CK3	CK4	CK5	CK6	CK7	CK8	CK9	CK10	CK11	CK12	CK13
I1	IF	ID	EX	MEM	WB								
I2		IF	ID	EX	MEM	WB							
I3			IF	ID	EX	MEM	WB						
I4				IF	ID	EX	MEM	WB					
I5					IF	ID	EX	MEM	WB				

CPU without forwarding

[illegible]

b) Rescheduled code:

CPU without forwarding

[illegible]

c) CPU with the forwarding paths

[illegible]

